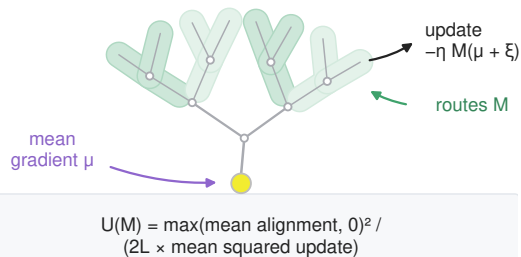
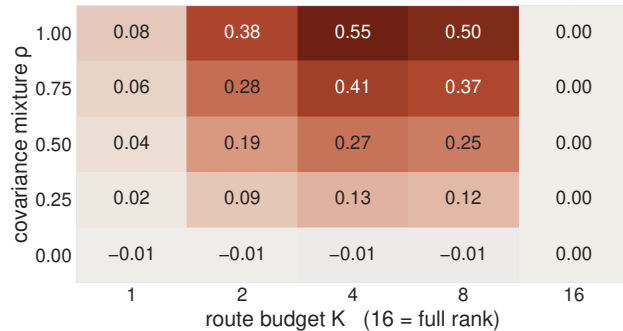


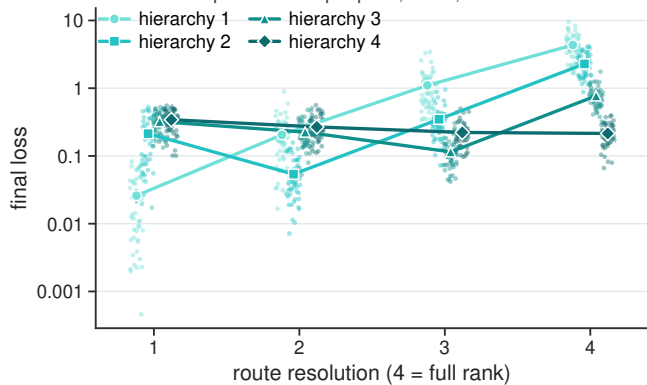
A

Fixed-operator special case

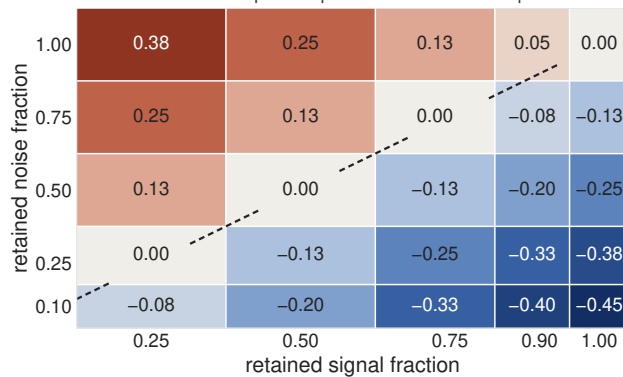
**B**Spectral capture advantage: subtree – random
mean of 50 paired seeds per cell**C**

Hierarchy × resolution

50 paired seeds per point; mean, 95 % CI within marker

**D**Projection boundary: Δ loss

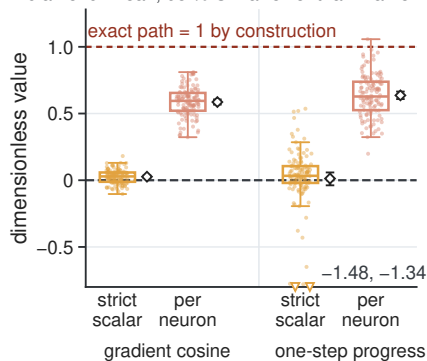
cells span midpoints between the sampled fractions

**E**

Trained checkpoints, one held-out batch

n = 120; box: median, quartiles, 1.5 IQR

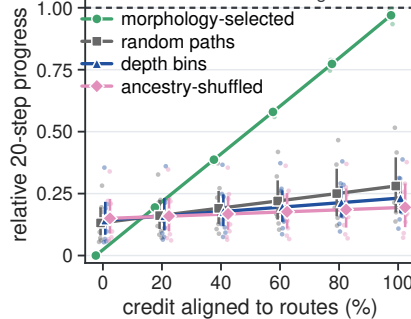
diamond: mean, 95 % CI narrower than marker

**F**

Iterative learning, eight arbors

n = 8 arbors per point; mean, 95 % CI

full gradient = 1

**G**

Rank–noise trade-off (analytic)

 $q^2 / (q + K\sigma^2)$, no data— $K = 1, q = 0.8$ --- $K = 2, q = 1$ 